

Clément Sarthe

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SUMMARY

Engineering student with experience in high-volume industrial environments and hands-on projects in hardware prototyping, Data Analysis, and AI applications. Eager to explore innovative fields like renewable energy, embedded systems, and robotics through a Master Thesis at Icam. Keen to gain deeper technical knowledge in Power Electronics, Battery Systems, and HIL Simulations with the Smart Grid project, alongside interests in drone-based technologies or industrial cobot programming and testing.

SKILLS

Prototype & Manufacturing: SMT, Laser Cutting, FDM 3D Printing, Composites

Design & Hardware: SolidWorks, KiCad, MATLAB/Simulink, Embedded Systems, Low Voltage Electrical Design

Software & Data Analysis: Python (NumPy, Pandas, Matplotlib), SQL, Web (HTML/CSS)

Quality & Project: Lean Six Sigma, PFMEA, Control Plan, Root Cause Analysis, Statistical Process Control

Languages: French (Native), English (Fluent), Portuguese & Spanish (Intermediate)

PROJECTS

Comma.ai Data Visualization: Python tool to visualize and analyze driving data from Comma.ai datasets using YOLO computer vision model and CAN data.

PCB Design: Custom PCB for temperature and humidity sensing with ESP32 microcontroller and Wi-Fi connectivity for Home Assistant integration.

Six Sigma Calculator: Python app to compute and visualize process capability indices and generate reports from CSV files.

Weather Station: Solar-powered, low-cost (<100€) station with temperature, wind speed, and air quality sensors enabling remote data access and off-grid operations.

AI Conveyor Belt: Arduino-based sorting system using stepper motors and AI vision to achieve real-time classification (Google Teachable Machine for training).

Night Light for Kids: 3D-printable LED lamp powered by 18650 cells and USB-C charging capability.

Tiny House: Led a 7-person team building a sustainable house on a tight budget, integrating passive solar heating and low-tech solutions.

RC Boat: Designed an RC boat with GPS live positioning, thrust and steering control over Wi-Fi.

EXPERIENCE

Production Quality Engineer (Apprenticeship)

Aug 2024 – Present

AUMOVIO (formerly Continental Automotive) – Toulouse, France

- Performed **Root Cause Analysis** on SMT lines by reviewing the process and quality data.
- Developed a full **tracking system** (database + web app) for 1500+ PCB magazines, improving traceability and inspections.
- Supported **IATF 16949** audit preparation by reviewing KPIs and strengthening our **QMS reliability**.
- Assisted **New Product Introduction** activities (PPAP), contributed to **PFMEA** and to **Control Plans**.
- Participated in **Gemba** walks and applied **Lean** actions to improve productivity and quality.
- Contributed to weekly **performance reviews** and quality improvement prioritization.

Business Analyst (Internship)

May 2024 – Aug 2024

Air France – Toulouse, France

- Wrote **functional specifications** and **acceptance criteria** for new software systems.
- Managed sprint backlogs and **facilitated** communication between stakeholders and developers.

Process Engineer (Internship)

May 2023 – Aug 2023

OGMA (Embraer Group) – Lisbon, Portugal

- Built an Excel VBA-based **SPC** system for electroplating aircraft parts, reducing manual inspection time.
- **Coordinated** daily maintenance follow-ups with Rolls-Royce.
- Supported **development** of maintenance capabilities for Pratt & Whitney GTF engines.
- Contributed to **PFMEA** for light-aircraft wing manufacturing with Daher.

EDUCATION

Icam, Engineering School – Toulouse, France

2020 – 2026

Master's Degree in General Engineering

Key courses: Industrial Systems, Computing, Embedded Systems, Lean Project Management, Supply Chain, IoT and Finance